Crime AI Analytics System

Geospatial Crime Analysis and AI-Assisted Dashboard

This report summarizes the analytical outputs generated by the Crime AI educational prototype.

1. Project Overview

The Crime AI project combines geospatial clustering, crime-type prediction, legal text similarity, computer-vision image comparison and a Flask-based web dashboard.

2. Crime Dataset Summary

Metric	Value
Total crime records	5
Detected spatial clusters	1
Records inside clusters	2

3. Crime Type Distribution

Crime Type	Number of Records
Theft	2
Robbery	1
Fraud	1
Assault	1

4. Geospatial Hotspot Analysis

DBSCAN clustering is used to identify groups of nearby geographic crime records. Cluster labels represent spatial groupings and should not be interpreted as proof that an area or person is dangerous.

Cluster	Latitude	Longitude	Records
0	28.61500	77.22500	2

5. Technologies Used

• Python

• Flask

• Pandas

• NumPy

• Scikit-learn

• DBSCAN

• Random Forest

• TF-IDF and cosine similarity

• OpenCV

• ReportLab

• Leaflet interactive mapping

6. Limitations

• The demonstration crime dataset is very small.

• Historical geographic clustering does not establish that future crime will occur at a location.

• Browser GPS provides current coordinates but does not provide a live crime-data feed.

• The crime prediction component is an educational prototype.

• Image comparison does not establish a person's identity.

• Legal similarity results are information-retrieval outputs and are not legal advice.

7. Future Scope

• Larger validated datasets.

• Time-based crime trend analysis.

• Interactive crime-category filters.

• Historical heatmaps.

• Improved model evaluation.

• Authorized live incident-data feeds.

• Privacy and security controls.

• Human review for consequential use.

8. Conclusion

The Crime AI prototype demonstrates how machine learning, geospatial analysis, natural-language processing and computer vision can be combined with a web dashboard to analyze a small crime dataset. The outputs are

intended for educational and analytical demonstration.